

Analysis First-Year Exam, August 2020, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let (a_n) and (b_n) be bounded real sequences. Define $\liminf a_n$ and $\limsup b_n$. Prove that

$$\liminf(a_n + b_n) \leq \liminf a_n + \limsup b_n \leq \limsup(a_n + b_n).$$

2. Let A be a subset of the metric space M . For each $t > 0$ let

$$(A)_t = \{x \in M : (\exists a \in A) \, d(x, a) < t\}$$

$$[A]_t = \{x \in M : (\exists a \in A) \, d(x, a) \leq t\}.$$

Is $\bigcap_{t>0} (A)_t$ equal to the closure of A ? Is $\bigcap_{t>0} [A]_t$ equal to the closure of A ? Proof or counterexample, as appropriate.

3. Let $f : [0, 1] \rightarrow [0, 1]$ be continuous and assume that for each positive integer n there exists $t \in [0, 1]$ such that $|f(t) - t| < 1/n$. Prove that there exists $a \in [0, 1]$ such that $f(a) = a$.
4. Let $f : X \rightarrow Y$ be a map between metric spaces, with graph

$$G = \{(x, y) : y = f(x)\} \subseteq X \times Y.$$

Prove that f is continuous if G is compact.

5. The function $f : (-1, 1) \rightarrow \mathbb{R}$ is differentiable and its derivative is bounded. Prove that there is a continuous function $F : [-1, 1] \rightarrow \mathbb{R}$ with the property that $f = F|_{(-1, 1)}$.